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Should Artificial Intelligence be incorporated into
the governing of Liberal Democracies?

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Abstract

Artificial Intelligence is a mechanism where a machine can learn from data and experience by writing, rewriting and thereby improving its algorithm towards a certain success. AI is expected to revolutionise all parts of life, including daily life, the economic sector and political system. However, the incorporation of AI bears certain risks. I will offer 2 arguments for why Artificial Intelligence should be incorporated into the governing of liberal democracies and how the menace could be minimised. First, I argue that AI is an extension of Collective Intelligence and that Collective Intelligence is the epistemic justification of liberal democracies. AI can make processes and society more effective and help to solve problems that are right now too difficult to solve. Secondly, AI will either way find its way into governing. If not done in an organised, transparent, explainable way under democratic measures, money and influence will decide who applies AI in government until possession of AI-applications determines who governs society. Without the incorporation of AI into governing democratically, AI will be a real threat to democratic structures. However, before one incorporates AI into governing, one needs to incorporate democratic measures into AI and guarantee to build algorithms that are human-centred and ethical and pursue the good of the people.

Keywords: Artificial Intelligence, Collective Intelligence, Liberal Democracy, Ethical Intelligence, Human-Centred AI, Machine Learning, Fairness, Bias, Transparency, Algorithms

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1 Introduction

Artificial Intelligence is a mechanism where a machine learns from data and experience by writing, rewriting and thereby improving its algorithm, to produce an outcome that optimises towards a certain success. Artificial Intelligence has experienced a remarkable evolution over the last decades. It led to the defeat of humans by machines in complex games such as chess or Go, transformed Google Search and made the generation of texts non-distinguishable from human-produced ones, possible through sophisticated natural language processing systems such as GPT-3. Applications of AI are steadily and presumably irrevocably approaching their way into industry, business, and the daily life of many human beings. Some say that AI will be the fourth industrial revolution, changing the way humans live, act and think forever [Chalmers, 2020].

The question of what amount of freedom and power one should give to AI and what the ethical boundaries of AI-applications should be is political. This debate is taken to another level when considering the incorporation of AI into governing itself. It is important to discuss this issue not just as a reaction to the evolution of AI, but to include predictions about what AI might be or could be in the future and how it could, should and will affect liberal democracy. With this thesis, I aim to make the connection between the considerably new theory of Collective Intelligence, the new philosophical field of AI and the technological opportunities and limitations. In the first section, I will define the terms AI (=Artificial Intelligence) and GoLD (=Governing of Liberal Democracy), as I will use them in the essay. I will start by discussing thoughts on individual, collective and artificial intelligence, followed by a description of the functioning, current and future capabilities of AI. The chapter will end with a definition of the term liberal democracy and I will present normative justifications for this form of governing, which I will relate to later on in the paper.

The second section debates AI as an extension of CI and argues that it hence should be applied to GoLD. This is followed by a discussion of the objections, including whether AI in GoLD will jeopardise “equality”. This part will lead to a limitation of the argument made before, claiming that AI-algorithms need to comply with democratic standards to be implemented in GoLD and that in most cases AI needs to be human-centred.

The third part argues that in the long term, AI will find its way into AI either way, so it is important to find a way of introducing AI that complies with democratic measures. For that, democratic measures need to be introduced into AI.

2 Section I: Collective Intelligence (CI), Artificial Intelligence (AI) and Governing of Liberal Democracies (GoLD)

2.1 Intelligence

Intelligence has been a field of study in biology, medicine, psychology, computer science and more. Definitions of the former usually included the term “mind” or “mental”, sticking intelligence to biology and the brain, a characteristic that belongs to a biological individual [Legg and Hutter, 2007]. Columbia Encyclopaedia describes intelligence as “the general mental ability involved in calculating, reasoning, perceiving relationships and analogies, learning quickly, storing and retrieving information, using language fluently, classifying, generalising, and adjusting to new situations” [Columbia University, 2006]. This defines intelligence based on the process of thinking.

To generalise the theory of intelligence, the “intelligent-agents-theory” gives a definition that is not based on the process of thinking, but on the capability of an agent to adapt to its environment and reach certain goals. A more detailed definition adds: Agents are intelligent if what they do is appropriate for their circumstances and goals if they are flexible to changing environments and changing goals if they learn from experience, and if they make appropriate choices given perceptual limitations and finite computation [Poole, Mackworth & Goebel 1998, p. 1].

An intelligent agent can thereby be a biological individual, a system of agents, a machine or anything one could imagine that can achieve goals in a wide range of environments [Legg and Hutter, 2007]. This includes for instance organisations, such as economic corporations; systems such as society and non-alive and “artificial” agents such as machines.

In this paper, I understand intelligence through the intelligent-agent-theory.

2.1.1 Collective Intelligence

CI is the concept of intelligence that does not apply to individuals, but groups of individuals or intelligent agents. It cannot just be described by the intelligence of the individuals but is found in the whole [Landemore, 2013]. The concept of collective wisdom is built upon the claim that the crowd outperforms the individual in decision-making, meaning it is not just a system working collectively in an intelligent way, but that this system is wiser than its individuals. This claim is made by James Surowiecki in “The wisdom of crowds”. Important is the phenomenon of diversity - described by Surowiecki as “diversity of opinion” or “each person should have some private information, even if it’s just an eccentric interpretation of the known facts” [Surowiecki 2005, p. 10].

Page argues that in problem-solving tasks, a group of humans that possesses cognitive diversity trumps ability. Cognitive Diversity means differences in thinking and interpreting, approaches for problem-solving, cognitive tools and perspectives [Page 2008, p. 7]. In a group, finding a solution cooperatively depends on the ability of the individuals to find solutions, but it depends to a greater extent on how different their approaches and tools are, such that they complement each other [Page 2008, p. 132]. The problem-solving-task is described as a mathematical function: there can exist local optima, which are the seemingly best solutions to a problem if one just looks at a specific interval. This local optimum can be achieved by a homogenous group of smart people or by a single expert. However, looking at the total function, there also exists the global optimum, which is the best solution for the task. This global optimum is likely to just be achieved if the whole interval of the function is considered. The whole interval can be translated into different approaches to problem-solving. So, it is indeed the cognitive diversity that leads a group to find the global, not just local, optimum.

Landemore constructs from here an epistemic justification of democracy, which I will discuss below.

2.1.2 Artificial Intelligence

Artificial Intelligence is the study and the design of intelligent agents. As said before, these intelligent agents can be anything that acts intelligent and hence AI is also the study of the intelligence of humankind.

The designed objects and the method of study are usually machines, and the term AI is often interchangeably used for intelligent machines, although this is just one part of AI. So when people talk about AI, what they usually mean is machine learning: machines that are capable of enhancing themselves through machine learning techniques. Their intelligence is “artificial” in the sense that it is not “natural”, but rather synthetic, meaning that it is not biologically but also not “fake” [Poole, Mackworth & Goebel, 1998].

In this paper, I understand Artificial Intelligence as the characteristics of machines to exercise tasks and reach goals in a way that they are not explicitly and initially programmed for and that they do this through machine learning techniques. I will use the words AI-model, ML-model and algorithm interchangeable, which means the sequence of instructions on how AI reaches its outcome. In passages, where I refer to AI as the whole branch of science about intelligent agents, I will explicitly say so.

2.2 Liberal Democracy

Democracy can broadly be defined as a method of collective decision-making of a group where during essential stages of the decision-making process the members experience equality in some way and where the decisions that are made in these collective processes are binding for all members of the group. Liberal democracy precisely refers to a political system which includes the following democratic measures: “the national government is accountable to the general population, and each individual is entitled to participate in the government directly or through representatives” [Bollen 1993, 1209], where citizens experience human and civic rights, where there is freedom of media and expression, universal suffrage, free and fair elections, separation of powers, more than one political party and distinct political parties, rule of law, fair and open society, right to private property and commonly a written constitution [Bollen, 1993].

Opposed to liberal are illiberal democracies where elections may be held, but where the true force of power is not transparent, where civil liberties are not present or violated and where no open society exists. Countries such as Russia or Turkey define themselves as democracies, but although elections are taking place, the overall system impedes a democratic change of power. The elements freedom of expression, freedom of media, equality, and fair society are crucial factors of liberal, but not necessarily illiberal democracies. These are the factors that enable collective intelligence and deliberation and these are the factors that AI might affect, as I will elaborate on. I draw my discussion also on the question if the incorporation of AI in liberal democracies, where the mentioned democratic measures are present and protected within the system, would change or endanger them. As in illiberal democracies, oppression of citizens is already taking place, AI is likely to worsen this oppression.

I use a broad interpretation of “liberal democracies”, which includes representative democracies, socio-democracy, republicanism but also other forms of liberal democracies, such as direct democracy, which could be enabled through new technological tools and ways of deliberation. I do so because as long as the democratic measures mentioned above are present and protected, the incorporation of AI could have very similar effects on all of these mentioned liberal democratic systems.

2.2.1 Governing of Liberal Democracies

When I talk about GoLD and the incorporation of AI, I include every process that leads to a form of decision-making of political significance. So in this paper, the term is used very broadly, including for instance information processing and news feeds, bureaucratic, administrative and executive tasks of and within the government, but also matters

connected to public debate, the judicial system, as well as very sophisticated processes of decision making such as developing and passing legislation.

2.3 Justification of Democracy

The epistemic approach argues that liberal democracy is morally favourable to other forms of governing because it produces good outcomes [Landemore, 2013]. “Good outcomes” are defined through the theory of CI: that the crowd is more likely to collectively make the right choice than an expert or a collection of experts and that people or intelligent agents that collaborate on a large-scale and a flexible manner are more intelligent together and produce better outcomes as individual intelligent agents.

Following the argument of Landemore, democracy is a collective decision method and is valuable as such, because it channels citizens’ collective wisdom/ “democratic reason.” She proposes that “the epistemic properties of democracy make it superior to dictatorship and any plausible variant of the rule of the few ” [2013]. Collective wisdom is here used for what was described as a collective intelligence being more intelligent than its members. The proposition is that a collectively governed society with collectively made decisions channels this collective wisdom and leads society to make better decisions than any individual or smaller group (even of the best minds) could. Cognitive diversity is the main reason for high collective intelligence. Page finds that a diverse group will outperform a sophisticated group in problem-solving. So, a diverse group will outperform a homogeneous group trying to do the same [2008].

Landemore argues that equality and freedom are grounds for public and inclusive deliberation - a deliberation in which all citizens can participate. This inclusive deliberation is supposed to “enlarge the pools of ideas and information, weed out the good arguments from the bad, lead to a consensus on the “better” or more “reasonable” solution” [2008]. She hypothesizes that “the advantage of involving large numbers” of people “is that it automatically ensures greater cognitive diversity” [2008, p.104].

Democracy is then justified because it reproduces “equality” and “liberty”, which enable citizens to possess cognitive diversity and democracy to produce better outcomes. By “equality” I mean the equal opportunity for every citizen to evolve an interpretation of the information they get and to participate in deliberation. By “equality” I also mean the one-person-one-vote scheme. Of course, for that the citizens need to also have the opportunity of receiving these information. By “freedom” I mean the possibility to evolve a particular, individual interpretation and to express this opinion.

3 Section II: AI, as an extension of CI, needs to be incorporated into Democracy

3.1 Argument: AI is an extension of CI.

..with CI being the justification of democracy, AI should be incorporated into GoLD. CI is driven

- i directly by cognitive diversity: the diversity of approaches that are channelled into an intelligent output
- ii indirectly by the size of input: meaning the greater the number of intelligent agents involved, the more probable is a high degree of cognitive diversity.

That means if AI can increase cognitive diversity and input, it is an extension of CI. The justification of democracy through the production of better outcomes under CI would demand that this type of AI is incorporated into GoLD.

AI can be an extension of CI by contributing to these factors in 3 different ways.

1. it can increase cognitive diversity through an AI-model which is inherently perfectly cognitive diverse.

The most obvious example of ML being a model with high inherent cognitive diversity is the ML-technique “Random Forest”. It is a variation of the “Bagging Tree”-method.¹ The input - data - is used to gain an outcome by the algorithm trying many different approaches, expressed as decision trees built on different bootstrap samples. They differ in the parameters they depend on and also in the way they are built. The prediction is determined by majority-vote² or by averaging³. taking into account all decision trees. This leads to the finding of the best solution through the “wisdom of crowds”, as described in the approach of Surowiecki. This averaging is better than having a single good decision tree - the equivalent of the single opinion of an expert.

This means that in tasks where there is an objectively right answer and where it is possible to model all possible cognitive diversity into the algorithm in a way it can read, then this algorithm will do the best possible job and could replace the human.⁴

¹The difference between the two is that for bagging at a node - points where tree splits in 2 - all of the available features are considered, whereas in “Random Forest” just the best feature for a split are selected.

²for tasks as classifications

³for tasks as regressions

⁴An example could be the calculation of limitation values for car uses in cities to keep air pollution below a certain level, a spam filter or text-correcting through AI.

This could be the case in task that are scientific, in a sense that they include a lot of factual data.

In cases that involve normative judgements its very questionable, if every information can be put into readable data for the algorithm and if every degree of cognitive diversity can be modelled.

2. it can increase cognitive diversity through AI-human-interactions, which adds another dimension to the cognitive diversity of a crowd. Also here the input number is increased by at least 1 additional intelligent agent.

One can increase the cognitive diversity of the system of intelligent agents - humans and machines- by adding a new dimension of diversity. Here, it does not replace the human, but acts as an advisor, as a tool, and as an additional provider of knowledge and judgement. To use this additional precision, humans need to understand the reasoning of the machine, so the decision can be criticised and shaped through deliberation.

3. it can increase the number of humans participating in large-scale deliberation, thereby increasing cognitive diversity and enabling the evolution of CI by giving needed the infrastructure.

Inclusive deliberation is often not feasible because of time and resources, AI could change that by enabling large-scale cooperation through technology, building infrastructure such that agents can communicate, deliberate, stay informed and adapt to new circumstances. Taiwan is for instance already applying a digital deliberation-platform called “vtaiwan”, which uses an AI-tool “Pol.is” to enable a discourse that is not driven by polarisation, but by finding common grounds [GOVLAB, 2020].

Not just in the case of deliberation, but for crowdsourcing in general AI can be an important tool to enable collaboration of humans that work for a common goal [Malone, 2015].

AI can be more effective than humans and conventional computer programs

The governing of liberal democracies includes a range of tasks that might be suitable for the use of AI-applications: the provision of information, administrative and bureaucratic tasks that can be automated, predictions that can be made about the future and trends until automated decision-making about laws. The technical advantages are that AI is

in some tasks fast, effective, less error-prone than humans, enables easy monitoring of processes and can process huge amounts of data.

These processes include tasks categorised under a) and general tasks that demand the processing of huge amounts of data, so tasks under 1., 2. and 3. In these processes, politics should apply AI, because it will be more effective in regards to money, time, resources and accuracy. These resources could be thereby saved and used for other needs.

An example is food production: Because AI can increase efficiency, it can increase agricultural productivity. AI could improve farming, by using computer vision to recognise pests or disease,⁵ by optimising nutrient input and water usage, by forecasting the weather and harvest time⁶ or designing gene mutated plants that are resistant and fruitful. It can also find new ways of food production as replacing meat with more environmentally friendly methods.⁷ The efficiency that could be gained through AI could also reduce trash, CO2-emissions and enable an in-total environmentally friendly way of living, without giving up comfort. In my broad definition of GoLD these are all processes that matter for the governing of liberal democracies.

AI may solve problems that are currently too difficult for humanity or society to be solved

This point builds on the previous one: AI does outperform and will increasingly outperform humans in certain tasks. In combination, this may enable the resolution of problems that governments are nowadays not able to solve. If AI can provide better outcomes than conventional governing and CI, it should be applied.

Here, the demand for AI also comes from the perspective that some problems that society is facing nowadays are so devastating that their non-resolution would endanger liberal democracies: an example is climate change which could lead to the resource scarcity of life-support goods, putting a threat to equality and human rights. If there is the chance that liberal democracies can - through AI - mitigate climate change and safely adapt to the part that cannot longer be stopped - they should at least try to do so.

Because AI can increase agricultural productivity, this can help to reduce hunger, despite a growing world population. Through efficiency in energy consumption and emissions it could mitigate climate change and adapt to the increasing number of natural disasters. This all will substantially help to reach the Sustainable Development Goals

⁵The Israelian firm “Prospera” enables efficient water and nutrient use on fields as well as identifies pest outbreaks on the fields through AI [Caine, 2020].

⁶The start-up Root AI is using AI-tools to monitor, when to harvest which plants [Caine, 2020].

⁷The Brazilian Firm “Fazenda Futuro” uses AI to analyze plants and design meat-alternatives out of this data [Caine, 2020].

[Caine, 2020].

3.2 Objection: AI in GoLD could decrease CI and jeopardise equality and liberty

equality and liberty Contrary to the arguments above, under some conditions, AI could also decrease cognitive diversity and CI, as I will elaborate on.

Equality and liberty are key conditions for inclusive deliberation, which is a key element of CI being the justification of democracy. If the application of AI into GoLD jeopardises these features, then AI does not increase cognitive diversity and is not an extension of CI, hence should not be incorporated into GoLD.

AI-usage could jeopardise equality and liberty in the following ways [Husted, 2021]:

- i AI-usage with good intentions but in a bad way such that it jeopardises freedom and equality
 - ii Ai-usage for profit, thereby putting profit over equality and freedom (see next section)
- Both depend highly on the design and training of AI, therefore I will discuss affiliated features of AI.

3.2.1 AI is unexplainable and untransparent

Humans can just comprehend AI-models if they are explainable and if they are transparent. Explainability refers to how well a model can be described such that humans understand this description. Complicated AI-models are often unexplainable because their algorithms are too long to be surveyed or too complex to be understood. This makes it impossible to understand how they obtained their outcome: which exact parameters were used under which weighting. This is especially true for Deep Neural Networks - AI-algorithms that consist of artificial neural networks with hidden layers.⁸ The way these hidden layers interact with each other is similar to the way neurons interact in our brain. In theory, “any AI, including black box neural networks can be converted to a large decision tree of nothing but “if” statements” - meaning that every single step that led to a decision could be captured-”, but it will only make it human-readable not human-understandable” [Yampolskiy 2019, p.6].

Transparency relates among others to the following points

- i is the model open to the public?

⁸This is called: Deep Learning. It is a type of machine learning where computing systems are used which designs were inspired by biological brains. Units or artificial neurons are built in multiple layers and are connected to interact and communicate with each other, as neurons are connected by synapses in the brain.

ii Are AI-applications marked as such?

iii Is revealed which data is used to obtain an outcome?

Transparency and Explainability are both conditions such that a human can comprehend the reasoning of the machine and its particular decisions. Equality between citizens demands understandability, such that outcomes can be questioned and errors corrected. For AI to increase the degree of cognitive diversity, it is equally important to understand arguments and reasoning of the machine, otherwise, its outcome is rather a shutting down debate, because the decision of the machine cannot be questioned.

However, it is indeed possible to design explainable AI-algorithms.⁹

One can distinguish between 2 types of explainability

1. models that are black boxes, but equipped with an additional tool to explain outputs post hoc. Global explainability refers to the ability to explain every decision-making-process: parameters that are in general considered important for the output, including their approximate weighting by the model.
2. models that are inherently interpretable: one can understand their parameters and the way they are used and weighted.

AI-researcher Cynthia Rudin emphasises in her work that black-box-explanations are insufficient in high stakes decision-making that affect individual members of society, such as in healthcare, criminal justice, because an accurate explanation cannot be guaranteed in every case.¹⁰

Rudin calls the trade-off between accuracy and explainability in inherently explainable AI a myth. Her research shows that inherently-explainable-AI needs to be less complex, but indeed it can have the same accuracy as very complex models. She hence emphasises using inherently-explainable-AI [Rudin, 2019].

This AI is important in the field of medicine, where relatively uncomplicated models with known parameters and weighting are used - for instance- to predict if a patient has cancer. They are used because in medicine it is often important to understand why a patient got a certain diagnosis. The AI-model may be reduced from 100 to the 6 most

⁹Albeit producing a less complex algorithm, the process of creating them is actually harder than creating complex algorithms, because one has to run multiple models in order to simplify the model while preserving accuracy.

¹⁰In explainable black-box-models one has to rely on the black-box-model as well as its explanation, which could be both erroneous. Rudin explains that if the explanation of the black-box-model would be correct in the sense that it would be correct in every case, then it would need to be identical with the model and hence complicated and ununderstandable in the same way as the original model. So the explanation has to be wrong, at least in some cases. This is unacceptable for high stakes decision-making [2019].

important parameters, which can be chosen by an AI itself, to be less complex but equally accurate. Examining the prediction of the model, the doctor can now decide if she wants to include parameters that are not included in the model, to make her final decision. As in medical diagnoses, it is important in processes concerning GoLD, to understand the reasoning and hence this type of AI should be applied and just this type would be valuable for decision-making.

However, in other tasks that do not involve the tracking and evaluation, AI might not need to be inherently, but still explainable.¹¹

Talking about transparency, the functioning and codes of ML-models are often not disclosed to the public, because they are protected by property rights of the company. In political processes, it is dangerous to extensively rely on models that are outsourced to companies and protected by property rights, because the interests or even stakeholders might not be clear.

Outsourcing decision-making to companies also opens up questions about accountability, which I will not discuss.¹²

In complex models that are not inherently explainable, it is also questionable if the public or anybody could understand the disclosed code. Revealing source-codes also makes the model vulnerable to hacking, because possible attackers gain technical insight and can observe weak spots.¹³

This shows that transparency is just valuable if a model is explainable in the first place. Transparency does not have to mean revealing the source-code, but disclosing the considered variables and the used data. In the case of doubt, the option is left to demand deeper insight into the code on a legal basis. If one wants to apply AI-models that are company-owned then there has to be a decision on how to value democratic rights in relation to the protection of intellectual property rights of the company. This involves moral judgements and the decision is ultimately a political one. The needed degree of explainability and transparency should depend on the degree that the model tracks and evaluates humans, because these processes are prone to privacy violations and discrimination.

Since AI-algorithms can influence (political) choice, for example by filtering content

¹¹to increase cognitive diversity

¹²not because they are not important but because there is no space for it in this paper.

¹³complex models as Deep Neural Networks bare another risk: in April 2022 scientist found that it is possible to plant undetectable backdoors in ML-models, giving possibilities to abuse [Goldwasser et. al., 2022]

and giving recommendations without the user knowing or acknowledging the influence (see next Section), It must be disclosed when and where AI is part of a tool and where it might drive decisions. People or institutions need to be aware that their data is collected, which data is collected and what of it is used for what.

3.2.2 ML-algorithms are not neutral math, but have embedded opinions. They are prone to discrimination and bias.

Algorithms or ML-models are far from being objective because they include a selection of parameters and weighting of the parameters that reflect the preferences and opinions of the training data and/or designer. Cathy O’Neil writes: ML-models are opinions embedded in mathematics [O’Neil, 2016]. Just as algorithms are not neutral,” their work is not impassive and apolitical. Algorithms search, collate, sort, categorise, group, match, analyse, profile, model, simulate, visualise and regulate people, processes and places” [Kitchin, 2017].

A key point is the definition of success - how is it defined if an algorithm works well and who defines it? Algorithms, meant to be used in GoLD, should have as success the common good to benefit society. They should be non-discriminative and fair.

Bias

describes the feature of a model that renders an unfair treatment towards a sensitive group or individual that is subject to the model’s decision-making. Fairness can in this regard be defined as the “absence of any prejudice or favouritism towards an individual or group based on their inherent or acquired characteristics” [Mehrabi 2019, p. 1].

can be embedded into a model in at least 2 distinct ways:¹⁴

- i through the data that is used to train the model.
- ii through the design of the algorithm: the algorithm can amplify or perpetuate bias that was introduced through the data, but it can also create bias itself.

Fairness

is an attempt to correct for the bias of an AI-model. Thereby a “sensitive group” is defined, that was found to be discriminated against in the original model. Fairness calibrates certain mathematical parameters of the output of the model so that they are more or less equal between the sensitive and non-sensitive group. Mathematical definitions of fairness are numerous and usually not compatible with each other. This

¹⁴A detailed list of many types of bias in ML, can be found in this paper: A Survey on Bias and Fairness in Machine Learning [N. Mehrabi, 2019]

might sound problematic, but in real life, we encounter the same issue: People could disagree on what they understand as fair, for example an equal distribution of a good or a distribution orientated on the particular need. The definition of fairness can already be a political statement:¹⁵

In general, which definition one perceives as fair will depend on the situation. Also, a combination of different definitions may be suitable for a model. However, these are not decisions that an engineer should make, especially when AI-models are used in GoLD.

Data

In most models, AI needs data to learn to construct its algorithm. The product is hence a reflection of that data.¹⁶

There are many¹⁷ examples of AI-applications that are found to be discriminatory against groups - usually groups that are already the most vulnerable in society. Parameters such as race or gender do not need to be explicitly in the model to generate bias or discrimination regarding that factor, since other factors or a combination of factors can be a proxy - an equivalent - to race or gender and thereby introduce the bias into the model.

Another problem is that often a huge amount of data of minority groups is not available and that the collection of this data bears correspondingly the risk of harm for these groups, for example by using the argument of needing to collect their data as an excuse to violate their privacy rights and surveille them.

If AI is created under these homogeneous circumstances, then it is not an extension of CI or cognitive diversity. This is problematic if decision makers rely on these models more strongly than on other intelligent agents of the society, for example if they trust these models without understanding them.

Kate Crawford is a researcher on AI-applications who focuses on the topics bias, discrimination and classification within AI. She writes that AI has a “white guy problem” [Crawford, 2016] - It is a problem that the creators of AI are mainly white, privileged

¹⁵An example on how fairness is perceived differently in real life is the debate about women quota in high rank management positions. One could perceive it as fair to give the position to the male applicant who has slightly better grades than his female competitor. Or one could argue for the employment of the female, because after centuries of gender discrimination it is still more difficult for women to get into these position. One could also argue that just in the case that both have the exact same qualifications (grades), the woman should be preferred. These are 3 opinions that can be related to 3 different definitions of fairness, which could all be reasonable, but which are not in detail compatible with each other.

¹⁶An example of how different an AI can evolve depending on its training data are the 2 microsoft-developed chatbots “XiaoIce” in China and “Tay” in the USA. Both bots had very similar initial conditions and learned from social-media-interactions with human users. Since 2014, “XiaoIce” has been an enormous success in China, with more than 40 Mio followers, from which many texted the bot saying “I love you”, while the USA-bot took very few hours to be racist, sexist and full of hate-speech [Neff, 2016].

¹⁷some examples are listed here

and male and that algorithms are trained with data in which other population groups are underrepresented.¹⁸ Crawford argues that it is not enough to simply add fairness calibrators to discriminative models. She emphasises thinking, not just of Ethics, but of Power relations and how AI will exercise power over citizens.

She argues that the algorithms are not just inherent technological mistakes, but that they are designed to discriminate. She proposes to exclude AI from all processes that track or evaluate humans [Crawford, 2022].

The bias of algorithms and the dependence on data that is prone to bias is a perturbing objection. It shows the need to think critically about algorithms and data. It can be met by having political regulations on how data can be used in a politically correct way as well as providing data for training that complies with democratic measure.

Furthermore, human judgements and decisions are based on data as well: things they learned, things they experienced and so on. Discovering that data is biased reveals a deeper insight than just saying that AI-models are unfair: it reveals bias that is present in society, in a subgroup of society, in the past and/or in the present. Data is often far from reflecting the whole society or democratic values: If it is collected from a racist and sexist subgroup, it will represent these features. AI, learning from this data will perpetuate the features into its algorithm.

It reveals where the representation of the society through the data does not comply with what the society democratically agreed to stand for - which in liberal democracies should be democratic values such as freedom and equality.

Argument: AI is not corrupt, sexist or racist if not made corrupt, sexist or racist

An advantage of artificial intelligence is that it is a systematic tool. AI will not favour a decision for personal reasons, because it has no personality. This is, of course, provided that corruption is not built into the algorithm.

With ML-models that are monitored, tracked, supervised and enhanced critically - preferable through CI - there is a good chance that they will likely be less unfair/biased than humans. Also, if ML-models are explainable and transparent, biased and (un)fairness are trackable more easily than in human decision-making [Hosangar, 2020].

¹⁸An example of the danger of homogeneity in tech and data is the hiring system of Amazon that was found to discriminate against female applicants. Amazon's "engineers concluded that the cause of bias was the data used for training the AI system, mostly consisting of resumes of male employees in reflection to the then trend of male dominance in the company and the tech industry" [Kodiyan, 2019].

3.2.3 Danger of Mass-Surveillance

If GoLD relies on AI that relies on data of the society, there is a threat that society will be mass-surveilled. Even if this is meant to be used for purely good purposes, it is a risk to society, because it could be misused by the government or other actors, or it could be stolen.

AI is already applied by authoritarian governments to mass surveillance their citizens. An example is Xinjian, where the Chinese government uses AI to suppress Uigers, the biggest oppression of an ethnic minority since the Third Reich. This mass-surveillance is possible through AI-technologies such as natural language processing and face recognition.¹⁹

The threat is also a reality for liberal democracies. In 2013 Edward Snowden leaked how the US-government was spying on its citizens, under democratic rule!²⁰ The disclosure showed how the government together with companies can violate privacy rights of the people through Big Data - AI nowadays can enable much more severe forms of surveillance.

The way liberal democracies, especially the USA, are treating their whistleblowers, does not cast a good light on the governments.²¹ Snowden, politically persecuted by the government of liberal democracy, had to flee to Russia.²²

These are all indications that trustworthy collection and usage of data by governments is not guaranteed in liberal democracies and that strong privacy rights are needed. In response to the massive campaigning of activists but even more the Black Lives Matter - movement, in 2020, Microsoft, IBM and Amazon announced to not sell the AI-technology “facial recognition” to law enforcement, because the risk of mass-surveillance, racial profiling and the violation of human rights through its usage are too high [Hamilton, 2020].

Covid, as the moment where everything went digital, was another turning point in the face of surveillance: surveillance of health and movement was suddenly a necessity for

¹⁹Human Rights Watch finds that authorities as police officers “use to communicate with the Integrated Joint Operations Platform , one of the main systems Chinese authorities use for mass surveillance in Xinjiang” through an app, which is aimed at “collecting personal information, reporting on activities or circumstances deemed suspicious, and prompting investigations of people the system flags as problematic.” In Xinjiang, but also in other parts, “the government has been blanketing the country with closed-circuit surveillance cameras (CCTV)”, which use AI to “automatically identify people from public surveillance footage streams and telephone calls” [Wang, 2019].

²⁰For instance, the documents revealed how the National Security Agency was “collecting the telephone records of millions of US customers of Verizon, one of America’s largest telecoms providers, under a top secret court order issued in April” 2013 [Greenwald, 2013].

²¹However in 2019 the European Commission put in law a guideline for the protection of whistleblowers [European Parliament, 2019].

²²At the time of writing this paper, Julian Assange is about to be sentenced in the US.

the common good.²³

The threat of mass surveillance needs to be addressed and It needs to be assured that this data is protected and exclusively used for what it is supposed to be used for.

Digital-strategy-consultant Francesca Bria worked, among others, for the Italian government and the city of Barcelona. She suggests that data collected from the people need to belong to the people [Bria, 2022]. Personal data, as well as metadata²⁴, cannot be owned by tech-companies. Data should be on the one hand protected for privacy, on the other hand, data that is used by politics and business, needs to be open also for other actors. The Data of the many cannot be the property of the few, because this creates monopolies in AI-technology.

3.3 Findings

There are processes in GoLD, where AI should not be incorporated, at least not for now. Before incorporating AI into democracy, democratic measures need to be incorporated into AI.

Despite having posthoc measures to correct and explain AI to some degree, this does not solve the full problem. In processes of GoLD that directly affect the life of individuals it is not enough to rectify posthoc, because then harm has already been done. It is not enough to implement fairness measures as a possibility to correct discrimination that is discovered by chance and due to attentive actors. It is also not enough to have explainable black-box-AI with explanations that are sometimes correct and sometimes not, because then there is no way to deeply question the decision-making. Hence, there are processes within GoLD that could be so devastating and harmful for citizens that AI should not be applied to, unless profound safety measures have been met, such as creating inherently explainable AI-models that act in a fair way, with a definition of fairness that was and is evaluated in an ethical way.

Crawford argues that AI-systems that track and evaluate people should not be applied to GoLD such as facial-recognition tools; “given the costly errors, discrimination and privacy invasions”. “Policymakers should not fund or deploy them until they have been vetted and strong protections have been put in place” [Kate Crawford 2019, p.1]. Algorithms that track and evaluate humans need to at least be inherently explainable and transparent.

²³In general, surveillance can also have good sights: for example prosecuting criminals of liberal democracies that carry out their crimes in their Thailand-summer holiday

²⁴Metadata is data about data, without revealing the data it is about. For example the metadata of a digital message would be its size, date or location of creation, but not the content of the message.

An example of how to regulate this is given in a proposal by the European Parliament, which classifies AI-applications into unacceptable, high, minimal or low risk, emphasising that AI cannot be used for the “unacceptable-risk”-category [European Parliament, 2021] and that certain measures have to be met in the high-risk category.

The discussion also showed that governments cannot be blindly trusted with data collection and technology usage, but that strong privacy rights and civil society are needed, to act as a strong part of GoLD to oversee the actions of the government. Also, the government needs to be transparent in regards to their AI-usage.

The objections have shown how AI should not be applied to GoLD. If algorithms track and evaluate people while being non-transparent, unexplainable, designed by a very homogenous group, owned by non-state actors and trained with biased data that is biased, then AI in GoLD will produce more harm than good. This also means that trustworthy data needs to be collected and made available so that companies have unbiased data to build their models on. But it also means that privacy has to be protected and that the data belongs to the people so that their data is just available for agents that care about the common good.

At this point it is fair to ask:

Why should AI be incorporated into GoLD after all, despite equality, deliberation and CI being at risk?

Continuing the discussion of this paragraph, the objections do not lead to the conclusion that AI should never be used, but that democracies need to be aware of the risk that AI and Big Data entail.

AI is simply not good enough yet to be applied to tasks that track and evaluate humans, because they are trained with data that reflects the word of the past and today-an imperfection. This shows how AI cannot replace humans, but be used as an additional model and tool that can increase cognitive diversity and find global optimum-solutions. To do that, humans need to understand its reasoning, which can be questioned and modified through CI. “Human-centred AI” exactly aims to produce this user-centred AI, that enhances “human performance in ways that make systems reliable, safe, and trustworthy. These systems also support human self-efficacy, encourage creativity, clarify responsibility, and facilitate social participation” [Shneiderman 2020, p.26:2].

So, AI-applications are not something to trust blindly, as one would trust a proven mathematical equation. Instead, algorithms are already political.

This identifies more processes to belong to the governing of liberal democracies, such as social media or lending. Liberal democracies need to be aware of the political implications that AI-application has on various fields. To protect equality and freedom and enable cognitive diversity, inclusive deliberation and CI, democratic measures need to be put into AI, by building transparent, explainable, accountable²⁵, non-discriminative models that are in the best case overseen by CI. Snow et al. for instance find that CI could for example also produce reliable datasets for AI to learn from, through crowdsourcing platforms [2008].

Fields and tasks of GoLD that do not involve the tracking and evaluation of humans are on the other hand suitable for broader AI-usage. AI-usage could help to fight climate change, natural catastrophes, poverty and hunger - by for instance analysing weather data and patterns to predict floods, tornadoes; mitigating forest fires and desertification or knowing where to plant which (modified) plant in order to feed the growing world population. In these cases, accuracy and complexity might be more important than human-centricity - case-dependent decisions that need to be made collectively, not by an engineer.

AI can reveal where the system of liberal democracy differs from its democratic values

As explained, in the introduction “AI” - and here I am using it not in the sense of just ML, but the broader scope, is originally the study of how we think. It reveals the way agents and systems think and what could ultimately happen if this way of thinking is put into practice on a large scale. AI eventually reflects the past and present of our society. Understanding that AI is trained with the past data of society ultimately gives insight into how the society thought and thinks and where trends, theories of thought and future actions might lead. Because AI represents what was fed into it through data and engineering, it can reveal bias and discrimination within liberal democracies. The examples of biased AI-models causing harm to vulnerable groups also show what happens to democracies if GoLD is unchecked and unquestioned.

So AI could help to identify where, how and why data and models are discriminatory

²⁵Accountability will not be discussed because it extends the scope of the paper, but it is an equally important part of AI-ethics and philosophy.

and how to overcome it. On the long run, these human-centred models could be enhanced, for example through the help of human CI, and ultimately be less discriminatory than any human, while being easily monitored and adjusted. Researchers such as Kartik Hosangar, argue that AI-models are already today much less discriminatory than humans - but this, of course, depends on the individual model [2020].

It is also a political decision if algorithms that have normative implications on society should reflect the world how it was, how it is, or how it should be (and then the question appears: how should the world be?). These are questions that should be answered under democratic measures, drawing on human CI. To make sure that AI really is an extension of CI and not replaces human participation, which ultimately would decrease cognitive diversity, a way of communication between machines and humans needs to be established. This, again, means that AI needs to be human-centred.

4 Section III: AI will find its way into governing anyways. To maintain liberal democracy, this needs to happen under democratic measures.

If AI is the fourth industrial revolution [Chalmers, 2020], then it will find its way into GoLD, similar to electricity or computers. Actually, AI already has been incorporated in GoLD, as I will elaborate on. This has happened and woö happen 1) by complying with democratic measures or 2) by not complying with them. AI that is applied in GoLD without legitimization is and continuously could be secreted, unevaluated and misused and thereby jeopardise democracy [Husted, 2021] in the following ways:

- iii Ai-usage for profit, thereby putting profit over equality and freedom
- iv applying AI for bad purposes

4.1 Argument: GoLD already relies on AI

The findings of the previous paragraph demonstrate that much more belongs to GoLD than just the government. Artificial intelligence as it exists today is a tool or a kind of software that is applied for tasks that demand a high level of data processing and interpretation. It is the next step in the evolution of data analysis and digitalisation. Many tools of data analysis are already used in information technology and incorporated into decision-making processes, hence AI is already used in GoLD. An example of this is news filtering and providing agencies such as Thomson Reuters. On its website, Thomson Reuters, the parent company of Reuters, declares that it has been injecting AI into its processes since the 1990s [Reuters, 2007].²⁶ Reuter’s filters also influence the filtering of other news agencies, even if they do not directly apply AI themselves. Luckily, Reuters is known to be an independent agency²⁷ which is said to comply with democratic measures.²⁸ Also in google search, machine learning is incorporated.process it. GoLD needs AI-applications to stay functional, also because of the amount of fake news that exists and that needs to be filtered from real news, for which AI is vital.

4.2 Argument: Actors that incorporate AI will become more powerful than liberal democracies that do not incorporate AI

If AI is not incorporated in GoLD then the chance is high that at some point in the future liberal democracies will be overturned by something that uses and masters AI.

²⁶Reuters benefited from implementing new technology quicker than others already in the early 20th century, when it incorporated radio to transmit news internationally [Reuters, 2007].

²⁷also guaranteed through the Reuters Trust principles

²⁸Of course, if Reuters were not trustworthy, it would not be such a successful news agency. In this case, one would find another news agency to rely on that is trustworthy. However, this is just possible given that there is no monopoly of data possession.

The alternative is that liberal democracies incorporate AI themselves for protection and efficiency. If a system is not efficient in comparison to others, then it is under the threat of erosion, because it could be overturned by systems that are more efficient and powerful.²⁹ This could happen through cyberwar if systems are not sufficiently protected, but it could also happen through slow but steady increasing dependence on AI systems that are not subject to regulations.

Algorithms can be persuasive and manipulate humans to have a certain opinion or to make a certain decision, thereby affecting GoLD.

Applications of AI could be troublesome because the way of filtering and spreading information may be misused to persuade voters and representatives in a manner desired by a concealed agent of interest. A certain way of spreading filtered information to targeted recipients to manipulate their opinions and decisions can corrupt elections by persuasion of voters through targeted ads and psychological tricks. Normal citizens as well representatives could be manipulated and pushed towards a certain view or decision.

An example of persuasion is the Cambridge-Analytical-scandal, which is said to have had a major influence on the 2016 US-presidential election, by using facebook-user-data to apply psychological targeting to influence the opinion of voters. Until today it is not exactly clear if they even used ML-techniques, but the case shows the power technology and data can have. A major problem was that there had not been any experience and knowledge about to what extent algorithms can influence the processes of GoLD.

This, again shows, how important transparency of AI-applications and Data-usage is, in terms of privacy but also to protect democracy. A better way is to apply GoLD transparently To prevent the misuse there needs to be an understanding of AI-technology

Tech-Industry

In 2020 the 5 biggest tech-firms³⁰ together had a market value that exceeded the one of Japan [Marker, 2021] - the third biggest economy in the world.³¹ The same companies are spending billions of dollars on AI-research [Rosenbush, 2022].

From the example of persuasive algorithms, it is clear that these AI-applications can have enormous impacts on GoLD. This will give increasing power to the algorithm-designers and -owners.

²⁹It's the economy, *.

³⁰Apple, Microsoft, Alphabet, Amazon, Facebook(Meta)

³¹Apple alone has a market capitalization that was just trumped by 7 countries in the world [Lishchuk, 2021].

The key question is, see the previous paragraph, who defines the success of a model? If AI-applications become increasingly powerful by influencing decision-making then there is a shift in the power balance between society and government, government and the providers /owners /commanders of the technology. The party that defines “success”, which parameters models include, and which data is used to train the model will have influence that does not comply with democratic measures. This broadly could be the tech-industry, but also actors that command from the shadows.³²

In 2021, whistleblower Frances Haugen released inner documents of Facebook, revealing how “the company’s internal culture prioritised profitability over its impact on the wider world” thereby operating “for the shareholders’ interest and not the public interest” [Waterson and Milmo, 2021]. Here, the definition of success is “profit” and putting “profit” over the public good. Profit is maximised if users are very active. Users maximise their engagement if the consumed content is very dramatic: content that is extreme and polarising [Ünver, 2018]. Haugen says: “much of the blame for the world’s increasingly polarised politics lay with social networks and the radicalising impact of services such as Facebook Groups” [Waterson and Milmo, 2021]. The 2021 US-Capitol-attack is just another example.

The power of the tech-industry primarily seems like an argument why AI should not be incorporated into GoLD: outsourcing processes of governing will give power that does not comply with democratic measures to tech-companies and outsourcing would be necessary in the first place because these companies own the AI-technology.

My argument is a different one:

It matters how AI is applied, in all processes, because, as shown in the previous section, all of the processes that incorporate the citizens data and influence their way of communication, information processing and deliberation, are ultimately political and part of GoLD. The emerging trend of economic tech-supremacy is unlikely to ever abate. AI will most probably accelerate this trend.

If not incorporated in a way that preserves equality, freedom, liberation and cognitive diversity, the companies that provide the AI-technology will choose who to equip or not

³²Corporate Europe Observatory and LobbyControl found in a study that in 2021 the amount of spending for lobbying to EU-Institutions enormously increases every year, already outreaching the lobbying of the pharma-, car- and financial sector. In one year they spend in total more than € 97 million for lobbying the EU, making it the biggest lobby-sector in the EU [Bank et al., 2021]. Also, the data they were able to use was at the point of publishing already 1-2 years old, current numbers will exceed those in the report.

equip with it. This will give advantages to those that are favoured by the providing actors. It is not very hard to imagine that this will drive into the situation of non-transparency, where the models will be owned by powerful stakeholders without the obligation to explain or share the function of their models and their interests. This option of manipulating decision-makers could shift power relations and jeopardise the principle of liberal democracy.

Due to that, liberal democracies need to step in and promote an alternative way of AI-usage. Society and decision-makers need to work closely with companies and technology, to instigate research toward the common good as success definition of the algorithm, to know, understand and regulate the industry in cases where regulation is needed (cases that track and evaluate humans) and to use its applications democratically and responsibly. Also, liberal democracies need to start doing this now³³, so that there is time and space for an organised and responsible introduction to processes of GoLD.

AI is also expected to change the labour-market: There are fears that companies will just replace humans by robots, which serves the “success” definition of the biggest profit. To inhibit mass unemployment and to make sure that everyone is better off when technology develops further, liberal democracies need to establish measures of distribution of welfare [Stiglitz and Kortinek, 2019]. Robots could as well fight the shortage of specialist in an aging society.

A way of counterbalancing the increasing power of tech-industry could be for instance the “The Windfall Clause”, a proposed distribution of extraordinary AI-profits for the common good. It is “an ex-ante commitment by AI firms to donate a significant amount of any eventual extremely large profits that they gained through breakthroughs in AI-capabilities to society” [O’Keefe et al., 2020]. Legally, this would be permissible, at least in US-American corporate law, as long as the expected costs at the time of the commitment are not too high. This shows that this type of regulation has to happen now, not after the “windfall”.

International Relations and Supremacy of AI-Leadership

If liberal democracies do not incorporate AI into their systems, AI will probably be incorporated - at some point in the future - by something other than liberal democratic systems. Or, to put it in Putin’s words “Whoever will be the leader in AI, will be the ruler of the world” [Allen, 2017]. Drawing on international politics and security issues, it does not need to be just the tech-industry that undermines democracy, but agents that

³³actually yesterday

precisely aim to harm democracy.

At least in security policies, the incorporation of AI is a need to protect the functioning of liberal democracies. The 24th of February 2022 has shown that wars of aggression are a matter of fact also for liberal democracies and not just stories from a history book.

The rise of technology, especially AI, is already changing warfare and global security. AI touches on some of the most important security issues of the modern world, which could be said to be cyber, nuclear power and propaganda.

The hacking of cybersystems, for instance, essential infrastructure, could paralyse whole cities/countries. It is not hard to imagine that in a city with a few million inhabitants the cutoff of electricity and water supply would cause the death of many people within days. Additionally, AI used within cyberattacks could be a very cheap and targeting way of warfare. One just would have to pay a small amount of very gifted IT-people a big amount of money to attack a city, organisation, country, but also specific individuals.

AI can also be misused to spread propaganda and fake news, as emphasised in the “persuasive algorithms” paragraph. In the war against Ukraine, videos appeared of both, Vladimir Putin declaring peace and Volodymyr Zelensky surrendering to Russia - both deepfakes [Wakefield, 2022].

Obviously, misinformation can find its way through more methods than deepfakes and in other circumstances than war.

This shows the new arms race that AI is causing. IT implies that liberal democracies have to incorporate AI into GoLD for security issues. Also, Cyber and propaganda attacks could be exercised by non-governmental actors, because, contrarily to nuclear power, AI could be available to any actor giving rise to new forms of terrorism and hacking. Despite these differences, the world history of nuclear weapons could give an idea of how to internationally regulate the security threat of AI that affects the whole world, as in nuclear weapons. However, liberal democracies will just have a say in the discussion about the regulation of AI-warfare if they play a role in AI at all.

These security concerns raise another question that does not just touch the field of warfare:

How can liberal democracies be sure that AI-ethics, which liberal democracies should consider very important, will also be important for the rest of the world if the AI-

technology of liberal democracy is not?

Liberal democracies have to stay a leader in AI-technology to determine their rules. Otherwise, the debate about how the technology should or shouldn't be used will not matter. This means that there have to be massive investments into all kinds of AI-applications in liberal democracies, as well as little regulation to not inhibit innovation. Liberal democracies should also preserve their digital sovereignty: the independence of their digital infrastructure, for example in the case of 5G.

4.3 Findings

There is a need to regulate AI-actors and tech-industry

The argument shows that there is a need to regulate AI-actors and the tech-industry in the present and future, or they will become a threat to democracy due to their power. AI-Applications should be introduced into GoLD in a slow, organised, supervised and monitored manner that is subject to democratic measures.

Whoever will take the lead in AI will determine the rules of AI

Neither persuasive algorithms nor security threats will disappear if one tries to keep AI out of GoLD. Contrarily: to keep "liberal democracies" as a form of governing, the incorporation of AI into liberal democracies and GoLD is indispensable for protective reasons. Citizens in democracies need to be warned of manipulative systems that may influence or alter their free choices and they need to be protected from propaganda and cyberwar. Awareness needs to be raised about the ways that data and technology can manipulate elections. Citizens and politicians will be influenced by AI even more in the future, this is why its existence, application and future evolution have to be acknowledged.

This means that AI is also about competition with other countries and systems. Liberal democracies need to heavily invest in AI-research and AI-applications that comply with democratic values, such that they exist and such that they are useful and functional compared to other approaches in the future, but also there have to be investments for AI as defense. Likely, AI-research for defense will be very little regulated and as in other times of history, research is also driven by arms race and warfare. In order to further this, innovation and freedom of innovation has to be guaranteed for the case of research and for applications that do not include the tracking and evaluation of humans.

5 Conclusion

The paper discusses the incorporation of AI into the governing of liberal democracies drawing on the epistemic justification of democracy through Collective Intelligence, which produces good outcomes. It argues that AI is an extension of CI under some circumstances and that in these cases AI should be incorporated into GoLD because it produces even better outcomes.

For problems that can be modelled sufficiently through data and to an extent that the highest degree of cognitive diversity is reached, AI-models should be used and might as well replace the human.

For problems that involve normative judgements such as the evaluation of humans or that involve the tracking of humans, AI-usage bears the risk of unfairness and discrimination. In cases, where normative judgements are involved, AI-applications should just be implemented after having been subjected to profound safety-measures that guarantee compliance with democratic measures, in the best way through the use of CI. This conditions that AI is human-centred.

But also, this AI reveals how societies or certain subgroups of society thought in the past and think in the present. It is hence a powerful tool to understand where theories of thought, trends and actions might lead the society.

However, to not repeat errors of the past, one has to evaluate carefully beforehand, how models are designed, what the training data represents and how “success” is defined in the algorithm. This means that before one incorporates AI into governing, one needs to incorporate democratic measures into AI and guarantee to build algorithms that have as their definition of success the good of the people.

The paper furthermore debates that AI will find its way into GoLD either way in the near or distant future. If not done in an organised, transparent, explainable, democratic way, money and influence will decide who applies AI in government until the point that possession of AI-applications determines who governs the government. The paper also underlines how the influence of AI-possessors is already increasing. Without the incorporation of AI into government democratically, AI will be an increasing threat to democratic structures. This is why democracies need to start creating AI that complies with democratic measures and introduce this type of AI in a slow and inspected way that leaves space for criticism, evaluation and improvement of the technology towards a usage that creates benefits for everyone. Democracies should start to do that as fast as they can because whoever will be the leader in AI-technology will also be able to determine its rules.

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